

Dye-sensitised solar cells were invented in 1991 by Professor Michael Grätzel of the Swiss Federal Institute of Technology, Lausanne, and coworker Brian C. O'Regan. Their work sparked a wave of dye-sensitised solar-cell research and led to the founding of several start-up companies.

Dye-sensitised solar cells have attracted much interest due to their low fabrication costs, relatively high

efficiency (especially under weak illumination), and ability to incorporate the dye colour employed. The textile form of these solar cells has several advantages over other types of solar cells, including light weight, high flexibility and mechanical robustness. Recently, the market demand for wearable devices has pushed interest in the development of high-efficiency textile-based dye-sensitised solar cells for energy suppli-

ers. The concept of textile-based DSSC is portrayed in Fig. 6.

Dye-sensitised solar cells also find application in building-integrated photovoltaics where solar panels can be built into various parts of a building's shell, including the rooftop. Silicon cells are preferred for rooftops as these are best for direct sunlight. However, dye-sensitised solar panels work well in diffused light, where silicon technology fails. For that reason, thin sheets of translucent dye-sensitised solar cells can be sandwiched between panes of glass, turning ordinary windows, skylights and glass facades into electricity generators.

Several companies are commercialising dye-sensitised solar cell technology, with Dongjin Semichem of South Korea and G24 Power Newport of the UK being the pioneers. Dongjin Semichem has the capacity to produce tens of thousands of modules every year. The aesthetic features of dye-sensitised solar cells are attracting architects, construction materials companies and window manufacturers. Dye-sensitised solar cells also power small electronic devices. One application gaining importance is the Internet of Things (IoT), which refers to a network of appliances, vehicles and other objects that are fitted with sensors and other electronic components to enable them to collect and transmit data.

G24 Power sells under the trade name GCell. It offers a dye-sensitised solar cell-powered Bluetooth-enabled wireless keyboard and plans to start shipping dye-sensitised solar cell-powered beacons that broadcast Bluetooth signals.

One demerit of dye-sensitised solar cells' early designs is that the liquid electrolyte, typically an organic solution of the iodide/triiodide (I^-/I_3^-) redox couple, is corrosive, volatile and prone to leaking. It can react with the dye, which reduces its long-term stability.

In year 2012, researchers at Northwestern University successfully replaced the liquid electrolyte with a novel semiconducting inorganic solid called fluorine-doped cesium tin iodide ($CsSn_{1.25}F_{0.75}$). It overcame the problem of corrosion and reactivity, and led to cells with conversion efficiencies of around 10 per cent.

To be concluded next month

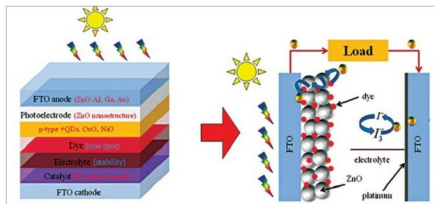


Fig. 5: Mechanism of dye-sensitised solar cells

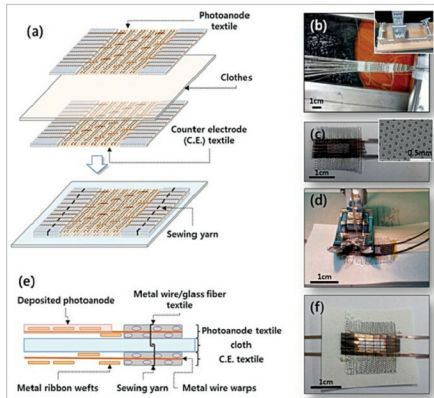


Fig. 6: (a) Fabrication concept for textile-based dye-sensitised solar cells made by sewing textile electrodes onto cloth or paper, (b) photograph of the loom (inset) used to weave textile electrodes, (c) a woven electrode (the inset shows an image of the stainless-steel ribbon with periodic holes), (d) sewing process to attach the woven electrodes to Hanji, a Korean traditional paper, (e) cross-sectional structure of the textile-based dye-sensitised solar cell prepared by sewing, and (f) a core-integrated textile-based dye-sensitised solar cell fabricated by sewing textile electrodes onto Hanji (www.openaire.eu)